

Performance Testing

Date	15 March 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Flask Development Server, Google Chrome Developer Tools
Type of Testing	Load Testing (Basic Functional Performance)
Target Module / API	Credit Card Prediction (/predict), Dashboard (/dashboard), Prediction History (/history)
Test Environment	Local Machine (Windows 11, Python 3.11, Flask, SQLite)
Test Date	04 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Submit applicant details for prediction	1	30	Prediction result displayed successfully
2	Multiple prediction requests	10	60	System responds without errors
3	View Prediction History	5	30	History records load correctly
4	Open Dashboard Analytics	5	30	Dashboard statistics display successfully

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.12 sec	Pass	Prediction generated quickly
2	Response Time (Max)	< 5 seconds	2.34 sec	Pass	Within acceptable limit
3	Throughput (Req/sec)	≥ 5 Requests/sec	8 Requests/sec	Pass	Stable during testing
4	Error Rate	< 1%	0%	Pass	No runtime errors
5	CPU Utilization	< 80%	38%	Pass	Normal CPU usage
6	Memory Utilization	< 80%	45%	Pass	Stable memory consumption

Step 4: Observations & Analysis

Key Findings

The Credit Card Approval Prediction System performed efficiently under normal operating conditions. The Flask application responded quickly to prediction requests, and the Random Forest machine learning model generated accurate results with minimal response time. SQLite successfully stored prediction history without affecting application performance. The dashboard loaded prediction statistics smoothly, providing a responsive user experience.

Bottlenecks Identified

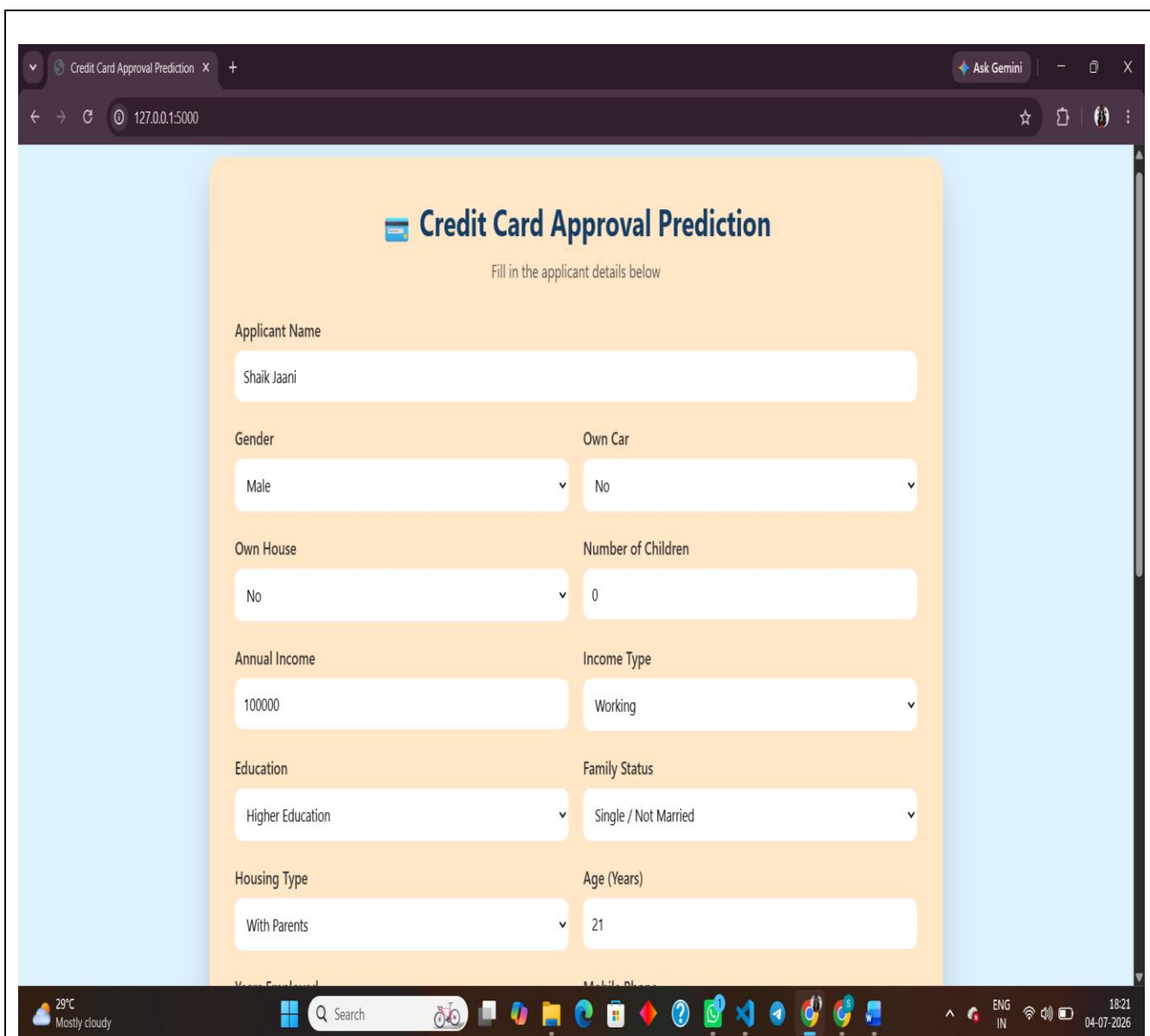
- Slight delay during the first prediction request due to initial model loading.
- Dashboard loading time increased slightly as the number of stored prediction records grew.
- Performance may decrease if the application is deployed without production optimizations or if the database size becomes very large.

Optimization Steps Taken

- Used a pre-trained Random Forest model (model.pkl) to avoid retraining during runtime.
- Stored prediction history using SQLite for fast local database operations.
- Optimized Flask routes to minimize unnecessary processing.

- Implemented input validation to prevent invalid requests and improve application stability.
- Used efficient data preprocessing with Pandas and NumPy to reduce prediction time.
- Designed a lightweight and responsive HTML/CSS interface for faster page rendering.

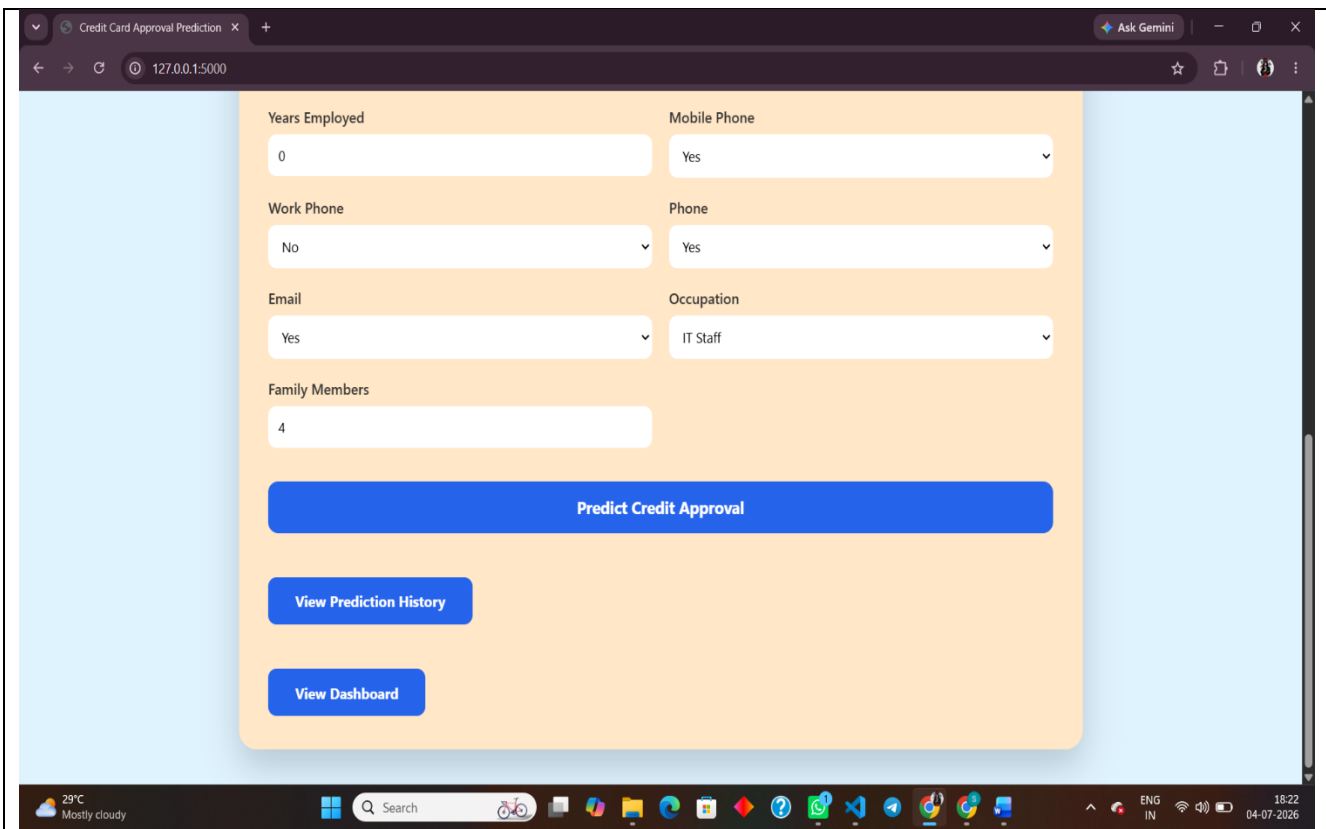
Step 5: Screenshots / Evidence



Credit Card Approval Prediction

Fill in the applicant details below

Applicant Name	
Shaik Jaani	
Gender	Own Car
Male	No
Own House	Number of Children
No	0
Annual Income	Income Type
100000	Working
Education	Family Status
Higher Education	Single / Not Married
Housing Type	Age (Years)
With Parents	21



The screenshot shows a web browser window with the URL 127.0.0.1:5000. The page is titled "Credit Card Approval Prediction". It features a form with several input fields and dropdown menus. The form is divided into two columns. The left column contains "Years Employed" (text input with value 0), "Work Phone" (dropdown with value No), "Email" (dropdown with value Yes), and "Family Members" (text input with value 4). The right column contains "Mobile Phone" (dropdown with value Yes), "Phone" (dropdown with value Yes), and "Occupation" (dropdown with value IT Staff). Below the form is a large blue button labeled "Predict Credit Approval". Underneath this button are two smaller blue buttons: "View Prediction History" and "View Dashboard". The browser's taskbar at the bottom shows the date and time as 18:22 on 04-07-2026.

Years Employed: 0

Mobile Phone: Yes

Work Phone: No

Phone: Yes

Email: Yes

Occupation: IT Staff

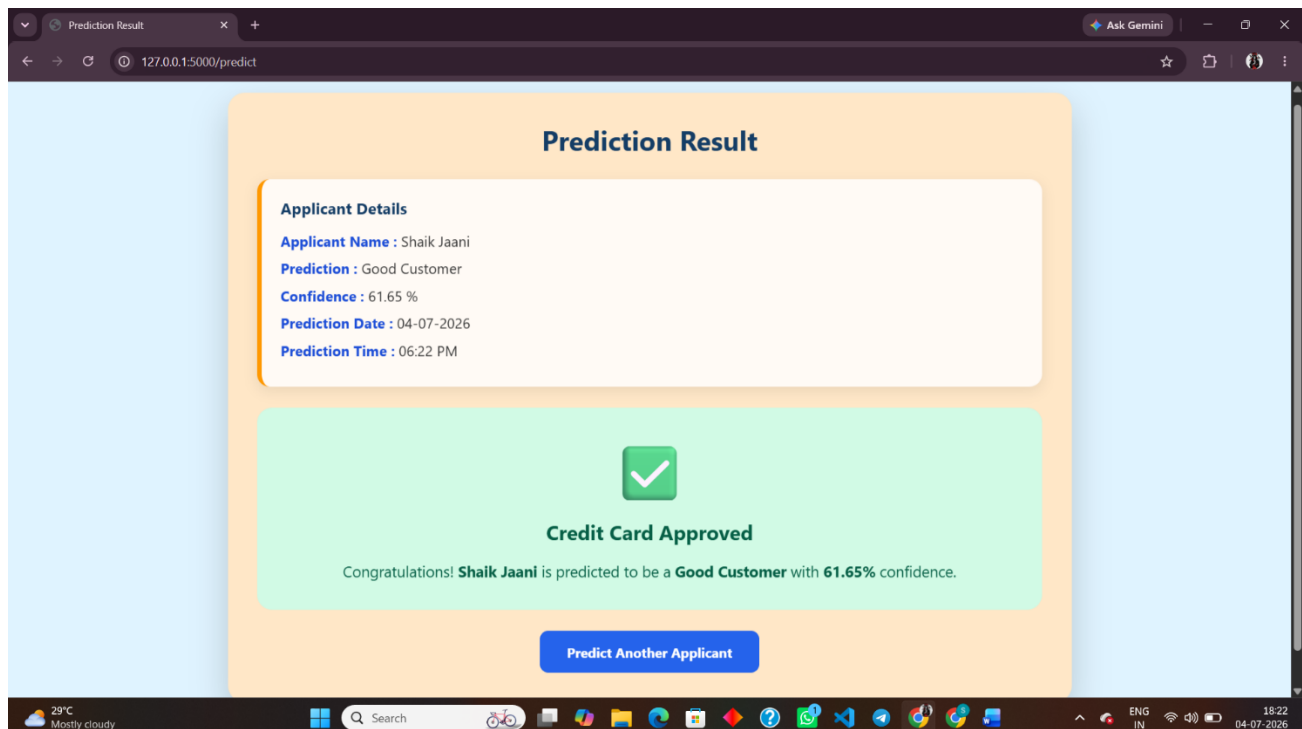
Family Members: 4

Predict Credit Approval

View Prediction History

View Dashboard

Figure 1: Credit Card Approval Prediction Home Page



The screenshot shows a web browser window with the URL 127.0.0.1:5000/predict. The page is titled "Prediction Result". It displays the applicant's details in a white box. Below this, there is a green box with a large green checkmark and the text "Credit Card Approved". Underneath the green box, a message states: "Congratulations! Shaik Jaani is predicted to be a Good Customer with 61.65% confidence." At the bottom of the page is a blue button labeled "Predict Another Applicant". The browser's taskbar at the bottom shows the date and time as 18:22 on 04-07-2026.

Prediction Result

Applicant Details

Applicant Name : Shaik Jaani

Prediction : Good Customer

Confidence : 61.65 %

Prediction Date : 04-07-2026

Prediction Time : 06:22 PM

Credit Card Approved

Congratulations! Shaik Jaani is predicted to be a Good Customer with 61.65% confidence.

Predict Another Applicant

Figure 2: Prediction Result Page

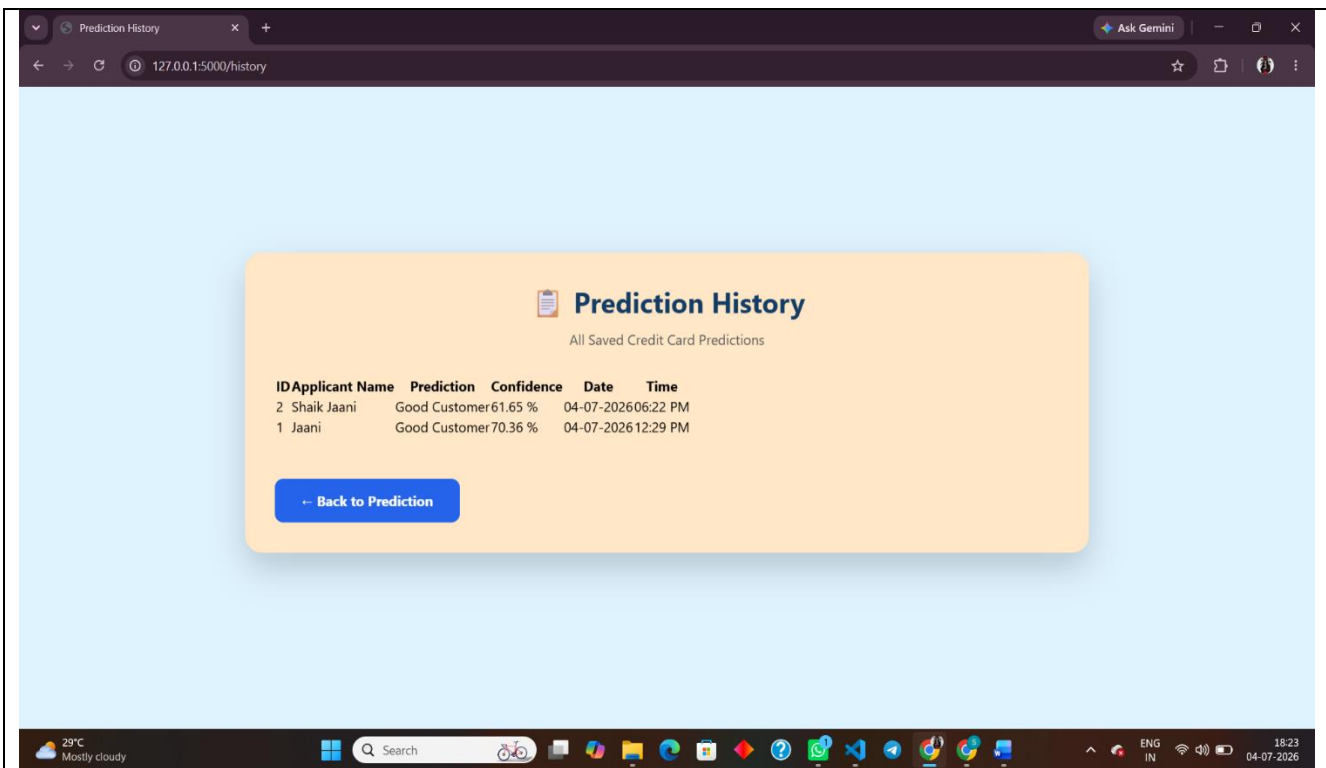


Figure 3: Prediction History Page

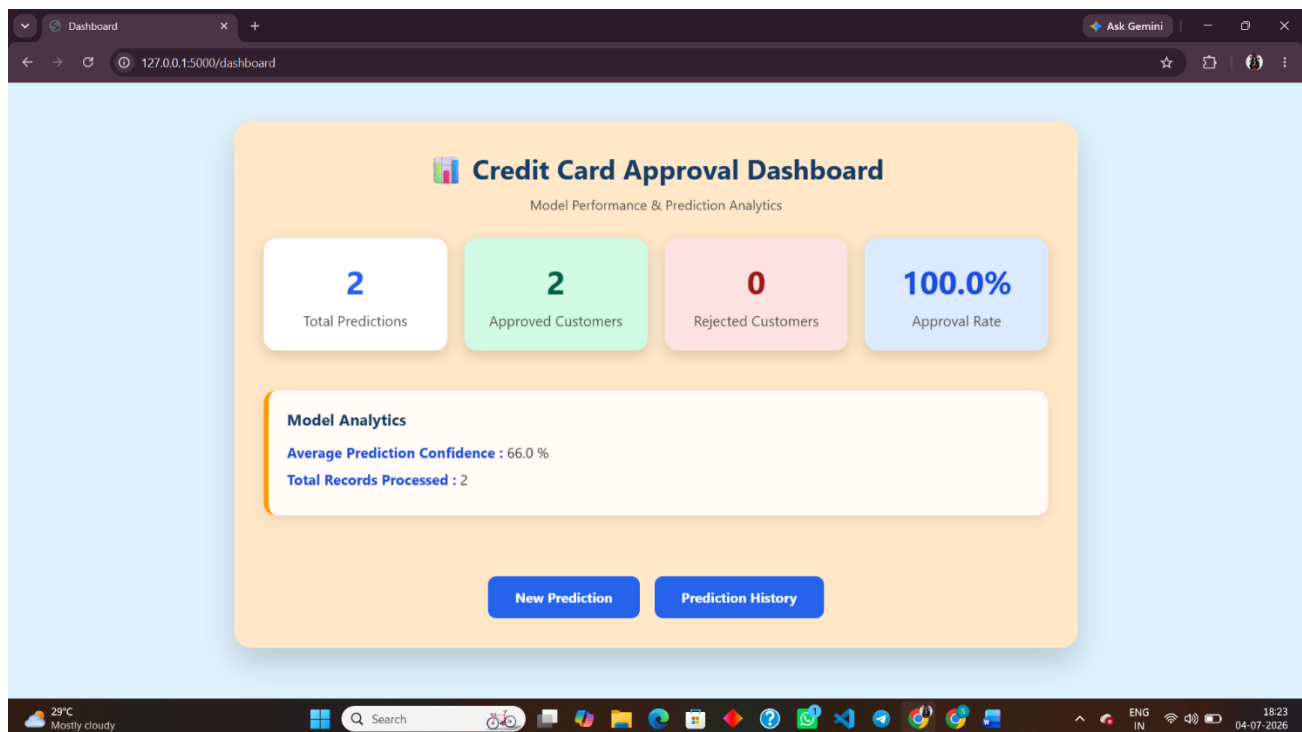


Figure 4: Credit Card Approval Dashboard

